

CMU-CS 462: Software Measurement and Analysis

Spring 2024-2025

Measuring External Product Attributes: Software Quality (1)

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12/01/2026

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Contents

1. Software quality
2. Software quality models: Boehm's model, McCall's model, ISO 9126 model, etc.
3. Software quality standards and metrics
4. Measuring customer satisfaction
5. Software quality assurance (SQA)
6. Software quality management



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Thảo luận

1. Chất lượng phần mềm là gì và tại sao nó khó đo lường hơn các sản phẩm vật lý?
2. Phân biệt "internal quality" và "external quality" trong phần mềm. Làm sao để đo được chúng?
3. ISO 9126 định nghĩa "Quality in Use" gồm những đặc tính nào? Đặc tính nào quan trọng nhất với bạn và vì sao?
4. Làm thế nào để xây dựng mô hình chất lượng phần mềm phù hợp với doanh nghiệp của bạn?
5. Đo lường mức độ hài lòng khách hàng có giúp cải thiện chất lượng phần mềm không? Nếu có, hãy đưa ví dụ.

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What is Quality?

■ Quality popular views:

- Something "good" but not quantifiable
- Something luxury and classy



■ Quality professional views:

- Conformance to requirement (Crosby, 1979)
- Fitness for use (Juran, 1970)



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Software Quality: Difficulties

- Need to account for “creativity” in the “design” of the product and the “requirements” rather than the product itself.
- Kind of art ...
- Which one has a higher quality?



Leonardo da Vinci

Mona Lisa (1479)



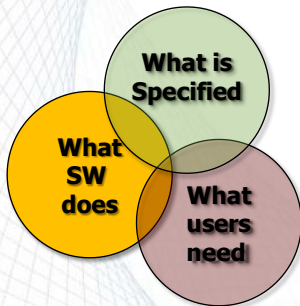
Pablo Picasso

Dorra Maar (1937)

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What is Software Quality?

/1



■ Conformance to requirement

- The requirements are clearly stated and the product must conform to it
- Any deviation from the requirements is regarded as a defect
- A good quality product **contains fewer defects**

■ Fitness for use

- Fit to user expectations: meet user's needs
- A good quality product **provides better user satisfaction**

Both → Dependable computing system

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What is Software Quality?

/2

Various viewpoints/perspectives:

- Conformance to customers' requirements
- Requirement, design and development quality
- Process quality vs. end-product quality
 - **Process quality:** higher usability and dependability
 - **End-product quality:** less failure
- Internal vs. external characteristics
 - **Relativity:** advantage over similar products
 - **Conformance:** conformance to standards

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Definition: Software Quality

- ISO 8402 definition of quality:
 - *The totality of features and characteristics of a product or a service that bear on its ability to satisfy stated or implied needs*
- ISO 9126 definition of quality:
 - *The totality of features and characteristics of a software product that bear on its ability to satisfy stated or implied needs (chịu đựng khả năng đáp ứng nhu cầu đã nêu hoặc ngụ ý)*

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Software Quality: Classification

- By observation:
 - Internal quality (quality while the product is being produced, including process and checks)
 - External quality (final product quality)
- By process:
 - Design quality
 - Implementation quality
 - Test quality
 - Maintenance quality

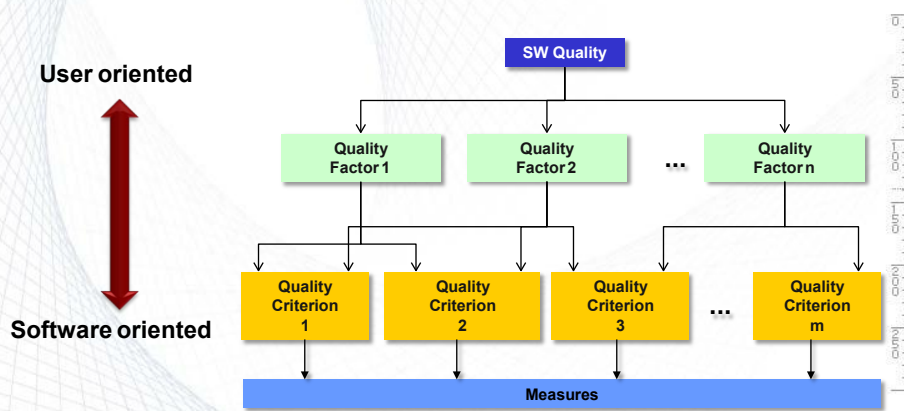
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Modeling Software Quality

- **Purpose:** To make the concept “quality” operational via refinement and measurement
 - Selecting some attributes (or factors)
 - Plotting relationship among attributes (many-to-many relationships)
- or
- Selecting some criteria (or intermediate and primitive constructs) to represent the attributes
 - Mapping criteria (or primitive constructs) to metrics

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Quality Model –Structure

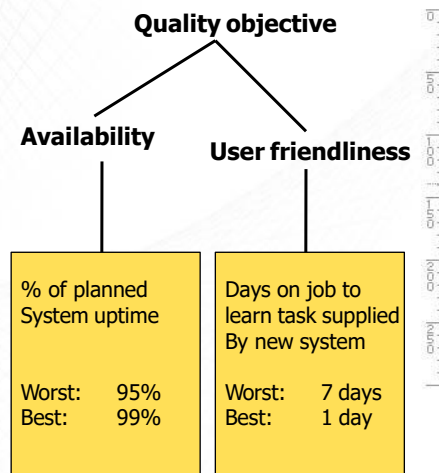


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Example: Attribute Expansion

■ Design by measurable objectives:

Incremental design is evaluated to check whether the goal for each increment was achieved.

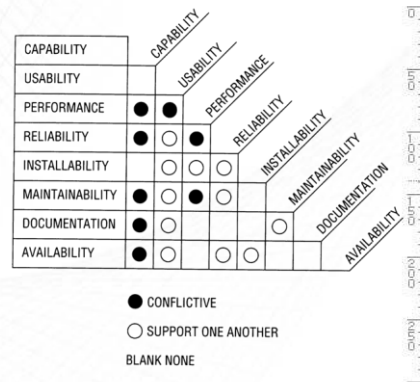


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Example: CUPRIMDA Model

■ Quality parameters (parameters for fitness):

- Capability
- Usability
- Performance
- Reliability
- Installability
- Maintainability
- Documentation
- Availability



Reference: S.H. Kan (1995)

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Example: Boehm's Model

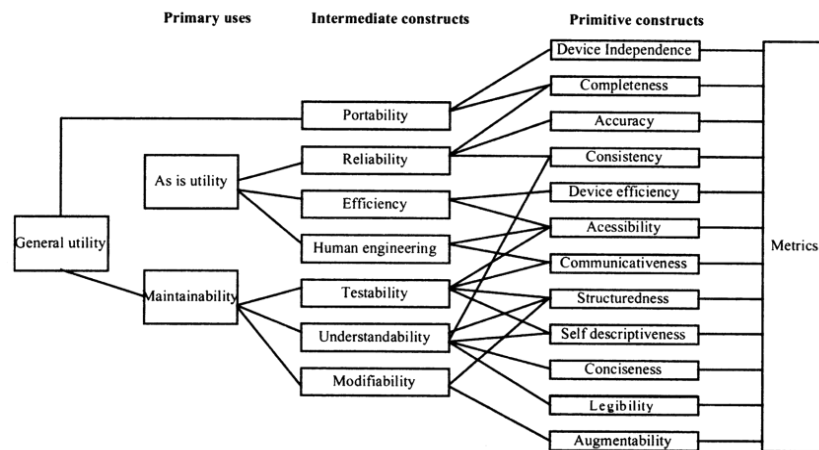


Figure from Fenton's Book

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Example: McCall's Model

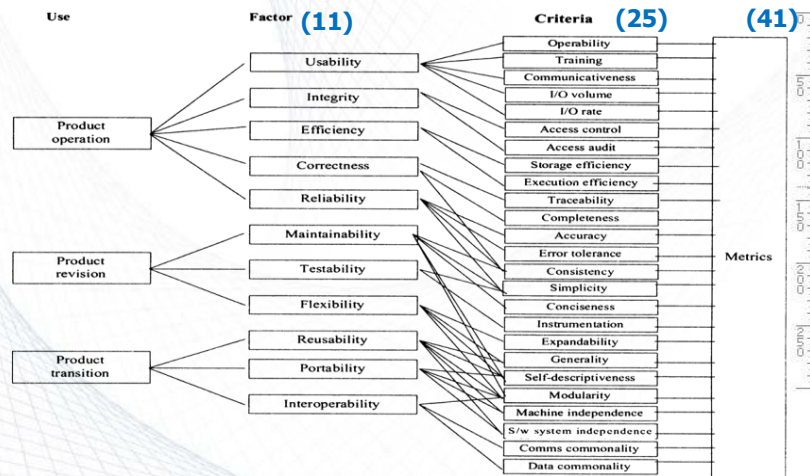
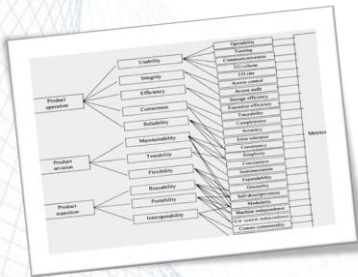


Figure from Fenton's Book

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McCall Model Elements



- McCall quality model is organized around three types of Elements:
 - **Quality Factors** (To specify): They describe the external view of the software, as viewed by the users.
 - **Quality Criteria** (To build): They describe the internal view of the software, as seen by the developer.
 - **Quality Metrics** (To control): They are defined and used to provide a scale and method for measurement.

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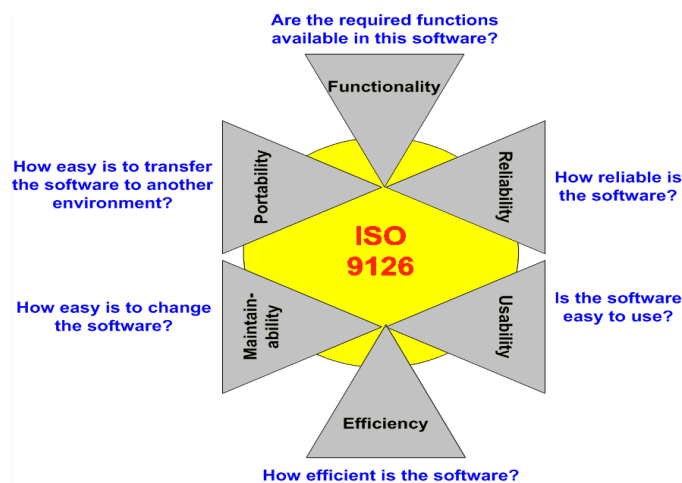
SATC* Software Quality Model

- *Software Assurance Technology Center (1996)
- NASA Goddard Space Flight Center

GOALS	ATTRIBUTES	METRICS
Requirements Quality	Ambiguity	Number of Weak Phrases
	Completeness	Number of Optional Phrases
	Understandability	Number of TBDs/TBAs
		Document Structure
		Readability Index
Volatility		Count of changes / Count of requirements
		Life cycle stage when change is made
Traceability		Number of software requirements not traced to system requirements
		Number of software requirements not traced to code and tests
Product(Code)Quality	Structure/Architecture	Logic complexity
		OOTD usage
		Size
Maintainability		Correlation of complexity/size
	Reusability	Correlation of complexity/size
Internal Documentation		Comment Percentage
	External Documentation	Readability Index
Implementation Effectivity	Resource Usage	Staff hours spent on life cycle activities
	Completion Rates	Task completions
		Planned task completions
Testing Effectivity	Correctness	Errors and criticality
		Time of finding of errors
		Time of error fixes
		Code Location of fault

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Quality Model: ISO9126



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Flexible Quality Model

- No need to stick to a fixed (published) model.
- You can define your own quality model based on users (customers and company) consensus. The model will be composed of quality attributes which are important for the given product.
- The model should be verified by actual measurement.

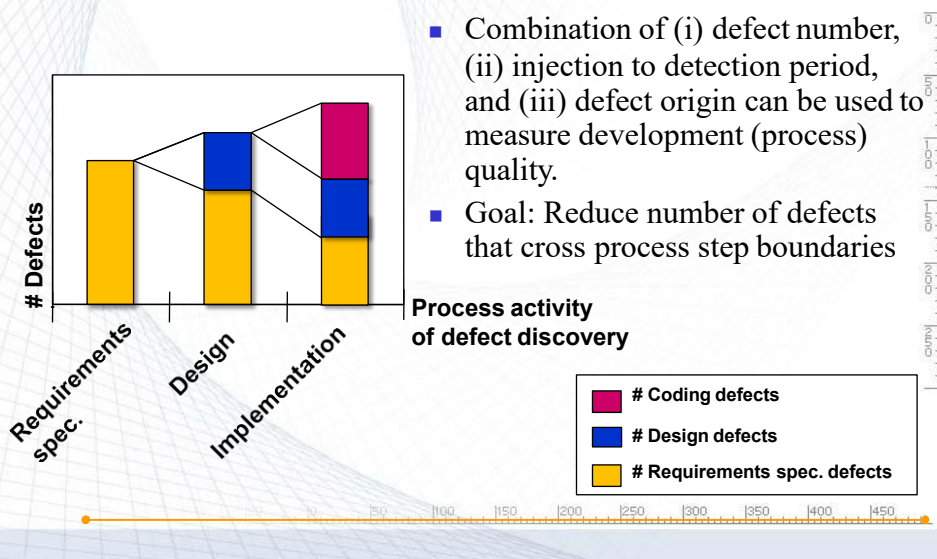
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Process-oriented Quality Models

- Purpose: To characterize process quality
- Measure process and product attributes
- Examples of Process Attributes:
 - Effort (person-months)
 - Duration (months)
- Examples of Product Attributes:
 - Code size (#classes)
 - Test set size (#test cases)
 - Correctness (#defects)

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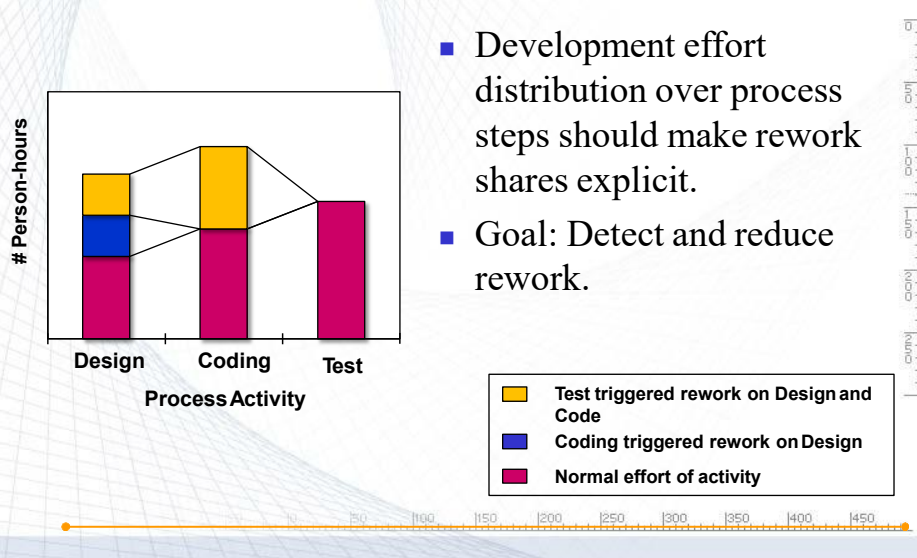
Example: Defect Distributions



- Combination of (i) defect number, (ii) injection to detection period, and (iii) defect origin can be used to measure development (process) quality.
- Goal: Reduce number of defects that cross process step boundaries

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Effort Distribution



- Development effort distribution over process steps should make rework shares explicit.
- Goal: Detect and reduce rework.

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Software Quality Standards

Software Quality Standards

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Software Quality Standards

- Software quality management standards (e.g., ISO 9000 series)
- Software process quality assessment standards (e.g., ISO15504 / SPICE)
- Software quality measurement standards (e.g., ISO 9126)
→ explained here

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ISO 9126: Parts

- ISO/IEC 9126-1:2001 Software engineering -- Product quality -- **Part 1: Quality model**
- ISO/IEC TR 9126-2:2003 Software engineering -- Product quality -- **Part 2: External metrics**
- ISO/IEC TR 9126-3:2003 Software engineering -- Product quality -- **Part 3: Internal metrics**
- ISO/IEC TR 9126-4:2004 Software engineering -- Product quality -- **Part 4: Quality in use metrics**

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Definition

■ Software Quality Characteristics

ISO 9126:

“A set of attributes of a software product by which its quality is described and evaluated. A software quality characteristic may be refined into multiple levels of sub-characteristics.”

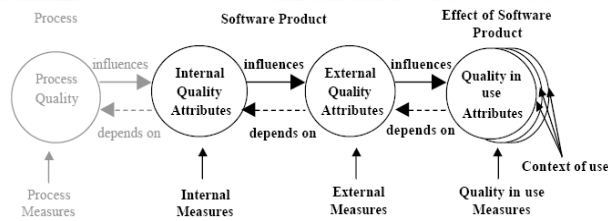
Một tập hợp các thuộc tính của một sản phẩm phần mềm mà theo đó chất lượng của nó được mô tả và đánh giá. Một đặc tính chất lượng phần mềm có thể được tinh chỉnh thành nhiều cấp độ của đặc trưng phụ.

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ISO 9126: Quality Lifecycle

- **Internal metrics** do not rely on software execution (→ static measures)
- **External metrics** are applied to running software
- **Quality-in-use metrics** are applied when the final product is used in real conditions

Ideally, internal quality determines external quality and external quality determines quality in use.



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ISO 9126 – Internal & External Base Measure

Internal Base Measures		External Base Measures	
Measure Name	Unit of Measurement	Measure Name	Unit of Measurement
34 Number of Functions	Function (number of)	38 Number of Functions	Function (number of)
35 Number of Data Items	Item (number of)	39 Operation Time	Minute
3 Number of Data Formats	Format (number of)	1 Number of Inaccurate Computations Encountered by Users	Case (number of)
4 Number of Interface Protocols	Protocol (number of)	4 Total Number of Data Formats	Format (number of)
5 Number of Access Types	Access-Type (number of)	5 Number of Abort Operations	Operation (number of)
6 Number of Access Controllability Requirements	Requirement (number of)	6 Number of Items Requiring Compliance	Item (number of)
7 Number of Instances of Data Corruption	Instance (number of)	7 Number of Interfaces Requiring Compliance	Interface (number of)
8 Number of Compliance Items	Item (number of)	8 Number of Faults	Fault (number of)
9 Number of Interfaces Requiring Compliance	Interface (number of)	9 Number of Failures	Failure (number of)
10 Number of Faults	Fault (number of)	10 Test Size	Byte
11 Number of Requirements	Requirement (number of)	11 Ease of Test Cases	Case (number of)
12 Number of Restorations	Restoration (number of)	12 Number of Restorations	Restoration (number of)
13 Number of Input Items Which Could Check for Valid Data	Item (number of)	13 Time to Repair	Minute
14 Number of Operations	Operation (number of)	14 Devia Time	Minute
15 Number of Messages Implemented	Message (number of)	15 Number of Restarts	Restart (number of)
16 Number of Interface Elements	Element (number of)	16 Number of Restorations Required	Restoration (number of)
17 Response Time	Second or Millisecond	17 Number of Testcases	Testcase (number of)
18 Turnaround Time	Second or Millisecond	18 Number of I/O Data Items	Item (number of)
19 I/O Utilization (Number of Buffers)	Buffer (number of)	19 Ease of Function Learning	Minute
20 Memory Utilization	Byte	20 Number of Faults	Fault (number of)
21 Number of Lines of Code Directly Related to System Calls	Line (number of)	21 Help Frequency	Access (number of)
22 Number of I/O Related Errors	Error (number of)	22 Error Correction	Minute
23 Number of Memory Related Errors	Error (number of)	23 Number of Screens or Forms	Screen (number of)
24 Number of Items Required to be Logged	Item (number of)	24 Number of User Errors or Changes	Error (number of)
25 Number of Modifications Made	Modification (number of)	25 Number of Attempts to Customize	Attempt (number of)
26 Number of Variables	Variable (number of)	26 Total Number of Usability Compliance Items Specified	Item (number of)
27 Number of Diagnostic Functions Required	Function (number of)	27 Response Time	Second or Millisecond
28 Number of Errors	Error (number of)	28 Number of Evaluations	Evaluation (number of)
29 Number of Built-in Test Functions Required	Function (number of)	29 Turnaround Time	Second or Millisecond
30 Number of Test Dependencies on Other Systems	Dependency (number of)	30 Task Time	Minute
31 Number of Diagnostic Checkpoints	Checkpoint (number of)	31 Number of I/O Related Errors	Error (number of)
32 Number of Data Structures	Data Structure (number of)	32 User Waiting Time of I/O Device Utilization	Second or Millisecond
33 Total Number of Setup Operations	Operation (number of)	33 Number of Memory Related Errors	Error (number of)
34 Number of Installation Steps	Step (number of)	34 Number of Transmission Related Errors	Error (number of)
		35 Transmission Capacity	Byte
		36 Number of Revised Versions	Version (number of)
		37 Number of Revised Versions	Version (number of)
		38 Putting User Feedbacks	Minute

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ISO 9126:Quality-in- Use

■ Effectiveness /

- The capability of the software product to enable users to achieve specified goals with accuracy and completeness in a specified context of use. *(Khả năng của sản phẩm phần mềm để cho phép người dùng đạt được các mục tiêu được chỉ định với độ chính xác và đầy đủ trong bối cảnh sử dụng được chỉ định.)*

■ Productivity

- The capability of the software product to enable users to expend appropriate amounts of resources in relation to the effectiveness achieved in a specified context of use. *(Khả năng của sản phẩm phần mềm để cho phép người dùng sử dụng lượng tài nguyên phù hợp liên quan đến hiệu quả đạt được trong bối cảnh sử dụng được chỉ định.)*

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ISO 9126:Quality-in- Use

■ Safety /

- The capability of the software product to achieve acceptable levels of risk of harm to people, business, software, property or the environment in a specified context of use.

■ Satisfaction /

- The capability of the software product to satisfy users in a specified context of use.

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ISO 9126:Quality-in- Use /2

Quality in use Base Measures		
	Measure Name	Unit of Measurement
1	Task Effectiveness	(a given weight)
2	Total Number of Tasks	Task (number of)
3	Task Time	Minute
4	Cost of the Task	Dollar
5	Help Time	Second
6	Error Time	Second
7	Search Time	Second
8	Number of Users	User (number of)
9	Total Number of People Potentially Affected by the System	Person (number of)
10	Total Number of Usage Situations	Situation (number of)

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Quality Model: ISO 9126

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1 : n relation
between
Characteristics
and
Attributes (Sub-
Characteristics)

Characteristics	Attributes		
Functionality	Suitability	Interoperability	Accuracy
	Compliance	Security	
Reliability	Maturity	Recoverability	Fault tolerance
	Crash frequency		
Usability	Understandability	Learnability	Operability
Efficiency	Time behaviour	Resource behaviour	
Maintainability	Analyzability	Stability	Changeability
	Testability		
Portability	Adaptability	Installability	Conformance
	Replacability		

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Life cycle

Previously

Withdrawn
ISO/IEC 9126-1:2001



Now

Withdrawn
ISO/IEC 25010:2011
Stage: 95.99 ▾



Revised by

Published
ISO/IEC 25002:2024

Published
ISO/IEC 25010:2023

Published
ISO/IEC 25019:2023

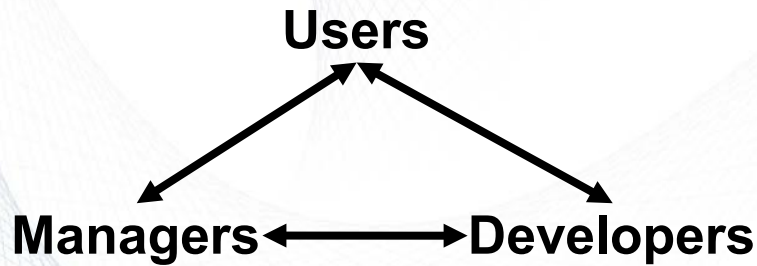
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ISO 25010:2023



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ISO 9126 –Views



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ISO 9126: Users' View

- Users are mainly interested in using the software, its performance and the effects of using the software.
- Users evaluate the software without knowing the internal aspects of the software, or how the software is developed.
- Users' questions may include:
 - Are the required functions available in the software?
 - How reliable is the software?
 - How efficient is the software?
 - Is the software easy to use?
 - How easy is to change this software?

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ISO 9126: Developers' View

- The process of development requires the user and the developer to use the same software quality characteristics, since they apply to requirement and acceptance.
- Since developers are responsible for producing software which will satisfy quality requirements they are interested in the intermediate product quality as well as in the final product quality.

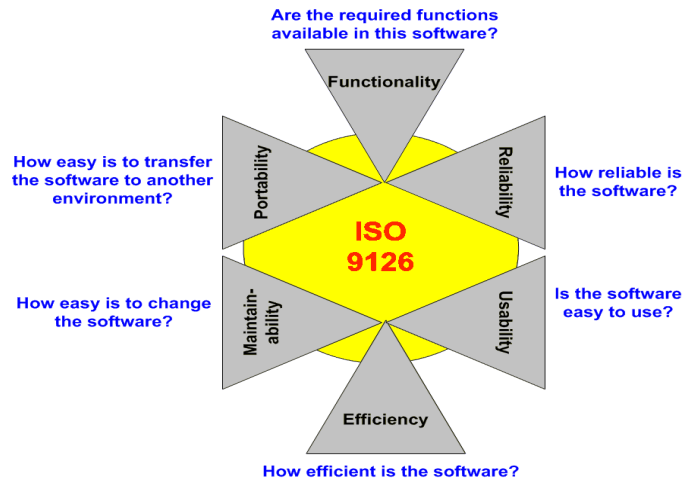
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ISO 9126: Managers' View

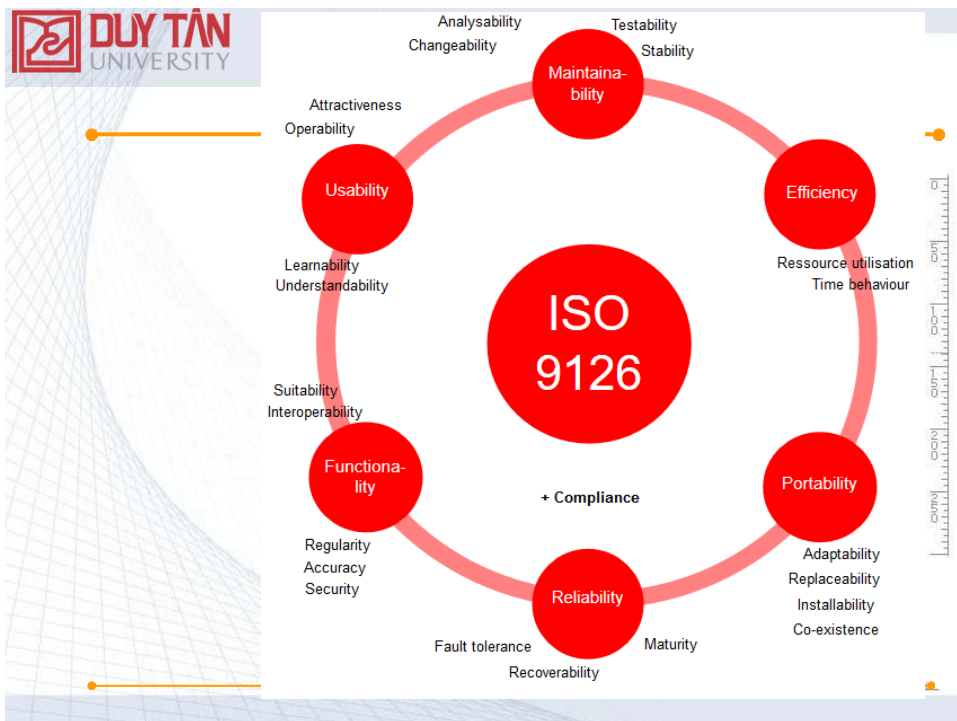
- A manager is typically more interested in the overall quality rather than in a specific quality characteristic, and for this reason will need to assign weights, reflecting business requirements, to the individual characteristics.
- The manager may also need to balance the quality improvement with management criteria such as schedule delay or cost overrun, because he/she wishes to optimise quality within limited cost, human resources and time-frame.

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Quality Model: ISO9126



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Quality Model: ISO9126

Characteristics	Attributes		
Functionality	Suitability	Interoperability	Accuracy
	Compliance	Security	
Reliability	Maturity	Recoverability	Fault tolerance
	Crash frequency		
Usability	Understandability	Learnability	Operability
Efficiency	Time behaviour	Resource behaviour	
Maintainability	Analyzability	Stability	Changeability
	Testability		
Portability	Adaptability	Installability	Conformance
	Replacability		

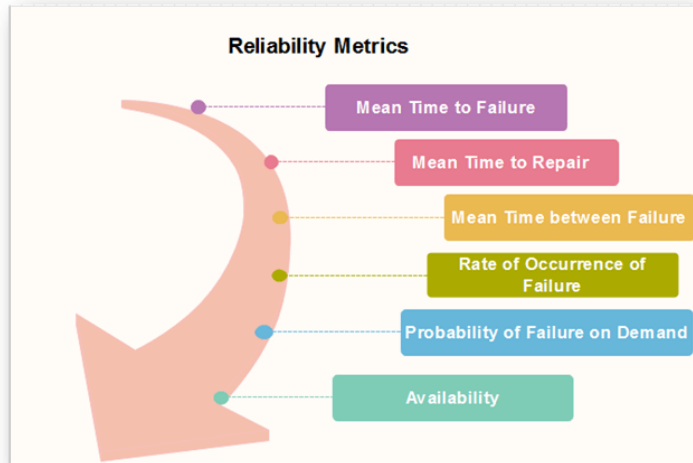
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ISO 9126: 1. Functionality

- **Suitability:** Attributes of software that bear on the presence and appropriateness of a set of functions for specified tasks.
- **Accuracy:** Attributes of software that bear on the provision of right or agreed results or effects.
- **Interoperability:** Attributes of software that bear on its ability to interact with specified systems.
- **Compliance:** Attributes of software that make the software adhere to application related standards or conventions or regulations in laws and similar prescriptions.
- **Security:** Attributes of software that bear on its ability to prevent unauthorized access, whether accidental or deliberate, to programs and data.

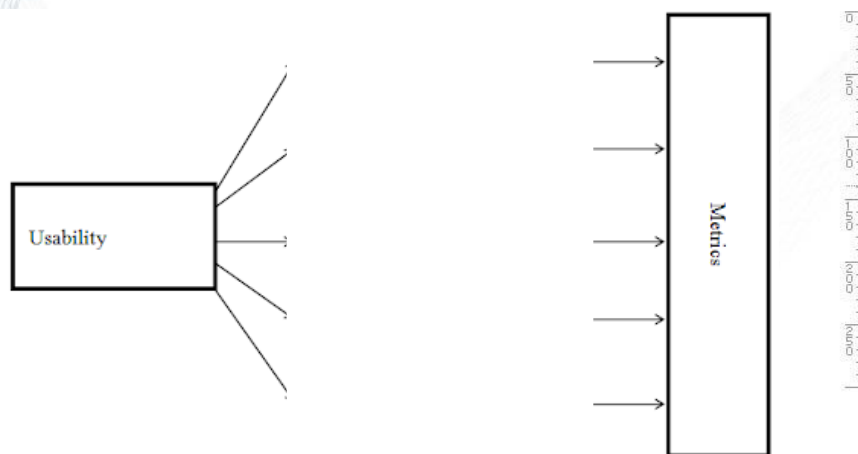
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ISO 9126: 2. Reliability



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ISO 9126: 3. Usability?



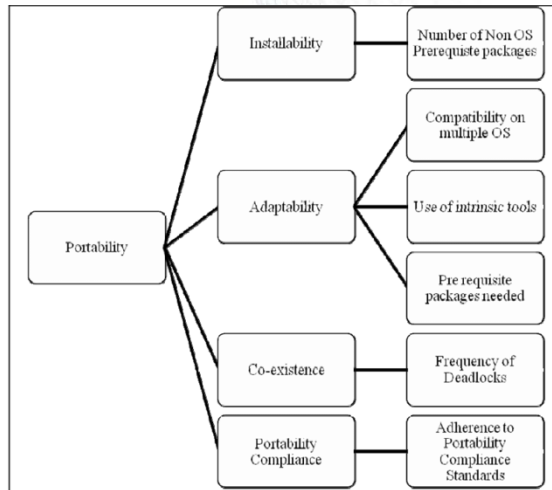
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- **Efficiency:** The extent to which a product or process can operate using the fewest possible resources.
- **Time behaviour:** Attributes of software that bear on response and processing times and on throughput rates in performing its function.
- **Resource behaviour:** Attributes of software that bear on the amount of resources used and the duration of such use in performing its function.

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- **Analysability:** Attributes of software that bear on the effort needed for diagnosis of deficiencies or causes of failures, or for identification of parts to be modified.
- **Changeability:** Attributes of software that bear on the effort needed for modification, fault removal or for environmental change.
- **Stability:** Attributes of software that bear on the risk of unexpected effect of modifications.
- **Testability:** Attributes of software that bear on the effort needed for validating the modified software.

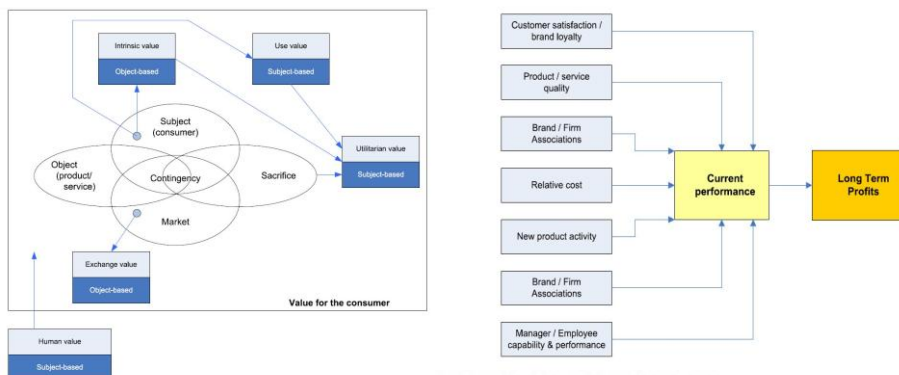
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Discussion in Group of 4 students

What is Customer satisfaction?
Customer satisfaction models?



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References and Assignment

Fenton, N., and Bieman, J. (2014). *Software Metrics: A Rigorous and Practical Approach 3rd*. Boca Raton FL, US: CRC press (442-471)
Kan, S. H. (2002). *Metrics and Models in Software Quality Engineering*. Addison-Wesley Longman Publishing Co., Inc.(426-437)

**Watch the video on the link and do the Quiz in this link.
Capture the result and upload the image on Assignment 10.**

<https://study.com/academy/lesson/measuring-customer-satisfaction-to-improve-retention.html#lesson>